

# Unidirectional LSTM Sequence Baseline

## Input Sequence Tensor

Shape:  $[B, T = 32, D = 117]$



## LayerNorm(117)

Standardizes skeletal landmark vector



## 2-Layer Unidirectional LSTM

hidden\_dim = 160 (Forward temporal flow)

num\_layers = 2 | Dropout = 0.3 |  $\approx 345\text{K}$  Params

Output:  $[B, T = 32, 160]$



## Temporal Global Average Pooling (GAP)

pooled =  $\frac{1}{T} \sum_{t=1}^T h_t \rightarrow [B, 160]$



## Two-Stage MLP Classifier Head

Dropout(0.3)  $\rightarrow$  Linear(160  $\rightarrow$  64)  $\rightarrow$  ReLU  
 $\rightarrow$  Dropout(0.3)  $\rightarrow$  Linear(64  $\rightarrow$  22)  $\rightarrow$  Softmax  
Output: 22 Exercise Class Probabilities  $P \in \Delta^{21}$